



GETTING STARTED

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<https://gitlab.com/cantankerous-impurity/Documentation/-/blob/core/GettingStarted.pdf>

How to make the code go...

Required dependencies

The easy ones...

- git
- make
- python3 with sympy
- C++ compiler with C++20 standard support [1]

Ubuntu 22.04 LTS commands

```
apt install make
apt install python3
apt install python3-pip
python3 -m pip install sympy
apt install build-essential
apt install gfortran
```

[1] In particular, support for [concepts and constraints](#) is required. Check [here](#) for a list of compliant compilers.

To make the code go... How?

Dependencies required

Some C and C++ libraries would be great

- boost
- sqlite3

Ubuntu 22.04 LTS commands

```
apt install libboost-dev  
apt install libsqlite3-dev
```

How make the code to go...

Necessary antecedents

Ubuntu 22.04 LTS commands

```
CUBA_DIR=<replace with host specific value...>
wget http://www.feynarts.de/cuba/Cuba-4.2.2.tar.gz
tar -xzvf Cuba-4.2.2.tar.gz
mv Cuba-4.2.2 ${CUBA_DIR}
cd ${CUBA_DIR}
./configure
make lib
```

The harder ones...

- cuba
- petsc
- slepc

```
PROJECT_NAME=cantankerous-impurity
```

```
PETSC_GIT=https://gitlab.com/petsc/petsc.git
PETSC_DIR=<replace with host specific value...>
git clone -b release ${PETSC_GIT} ${PETSC_DIR}
PETSC_ARCH=arch-${PROJECT_NAME}
PETSC_ARCH_DIR=${PETSC_DIR}/${PETSC_ARCH}
cd ${PETSC_DIR}
./configure \
    --download-mpich --download-fblaslapack --with-scalar-type=complex
make all
```

```
SLEPC_GIT=https://gitlab.com/slepc/slepc.git
SLEPC_DIR=<replace with host specific value...>
git clone -b release ${SLEPC_GIT} ${SLEPC_DIR}
cd ${SLEPC_DIR}
SLEPC_ARCH_DIR=${SLEPC_DIR}/${PETSC_ARCH}
./configure
make all
```

Code how go to make the

Deepen dent seas wreck wired

Ubuntu 22.04 LTS commands

```
PROJECT_DIR=<replace with host specific value...>  
mkdir ${PROJECT_DIR}
```

```
PRECALC_DATA_DIR=${PROJECT_DIR}/precalc-coeffs  
mkdir ${PRECALC_DATA_DIR}
```

```
PRECALC_SUBPROJECT=Precalculation  
PRECALC_DIR=${PROJECT_DIR}/${PRECALC_SUBPROJECT}  
PRECALC_SPEC=${PROJECT_NAME}/${PRECALC_SUBPROJECT}  
PRECALC_GIT=https://gitlab.com/${PRECALC_SPEC}.git  
git clone ${PRECALC_GIT} ${PRECALC_DIR}
```

```
COMPUTATION_SUBPROJECT=Computation  
COMPUTE_DIR=${PROJECT_DIR}/${COMPUTATION_SUBPROJECT}  
COMPUTE_SPEC=${PROJECT_NAME}/${COMPUTATION_SUBPROJECT}  
COMPUTE_GIT=https://gitlab.com/${COMPUTE_SPEC}.git  
git clone ${COMPUTE_GIT} ${COMPUTE_DIR}
```

These work... as long as the dependencies have been correctly installed and configured...